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KHAZZANI

SMART WATER TANK

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ABSTRACT

One of the most important challenges of this era is the scarcity of water availability in some countries around the world, an example of which is the Hashemite Kingdom of Jordan.

In Jordan, we face a shortage of water sources, where the per capita share of water is currently about 63 cubic meters annually, which is much lower than the global water poverty line estimated at 500 cubic meters per person annually.

To address these challenges, the government is working on implementing strategic projects such as the National Carrier Project, expanding the use of treated water, and **improving water resource management**, with the aim of ensuring the sustainability of water supply for citizens in the future.

One of the projects aimed at improving water resource management is the Smart Water Tank project.

CH1: INTRODUCTION

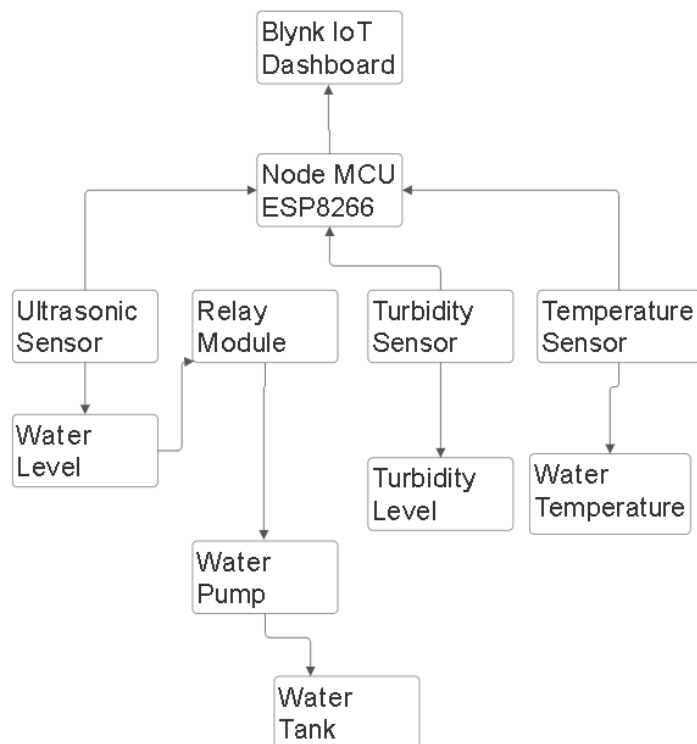
Water is the main element used in daily living, whether for home or commercial purposes. Overuse of water is now a significant problem that is harming our environment. Lack of water may result from excessive wasting, which can also cause other environmental issues like climate change, droughts, rising pollution, and rising human demand. As there is a limited supply of fresh water, it is crucial to use and manage it properly. Monitoring water waste in different sectors, such as residential, commercial, or industrial locations, is urgently necessary.

CH2: PROPOSED SYSTEM

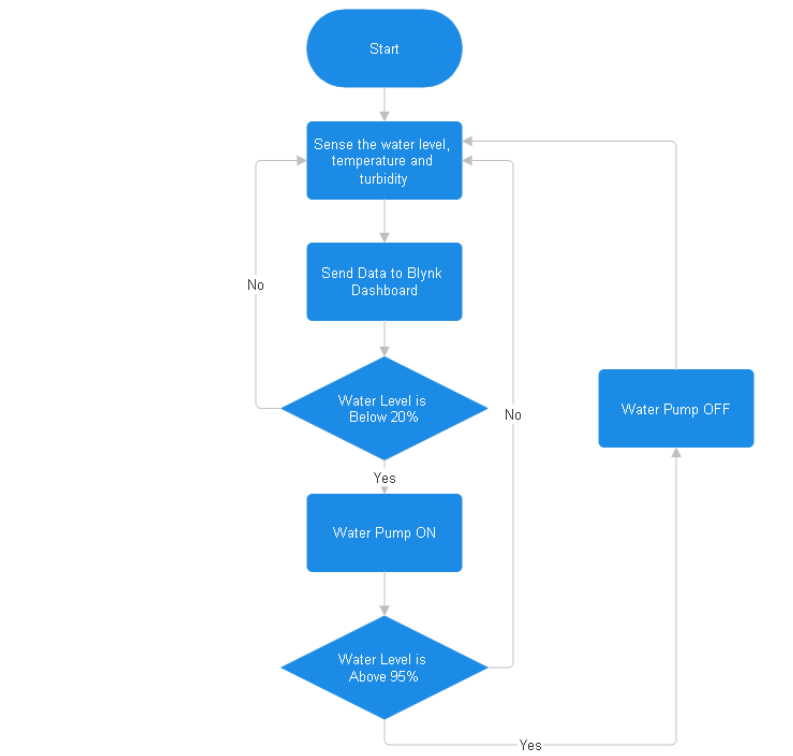
The Proposed System is named as KHAZZANI.

The components that are being used in the proposed system are NodeMCU ESP8266 Microcontroller, 5v relay module for water pump control, Ultrasonic for water level calculate, temperature sensor, turbidity sensor, water pump and AC to DC Power supply 12v for the water pump.

2.1 Block Diagram of Proposed System



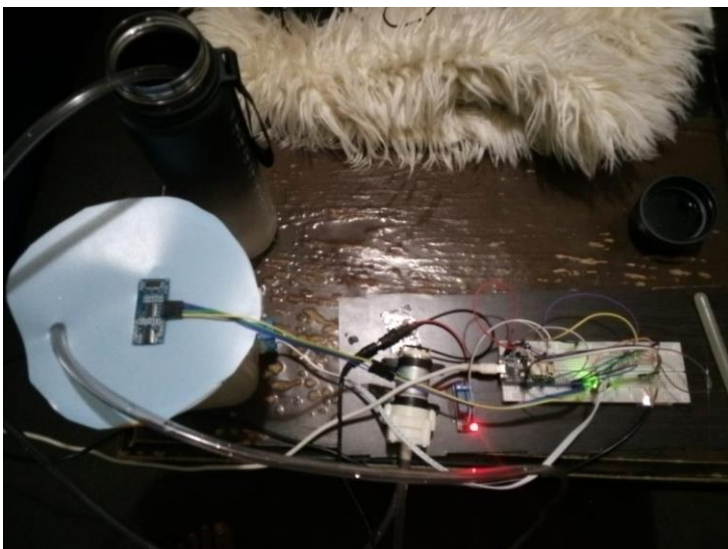
2.2 FlowChart of Proposed System



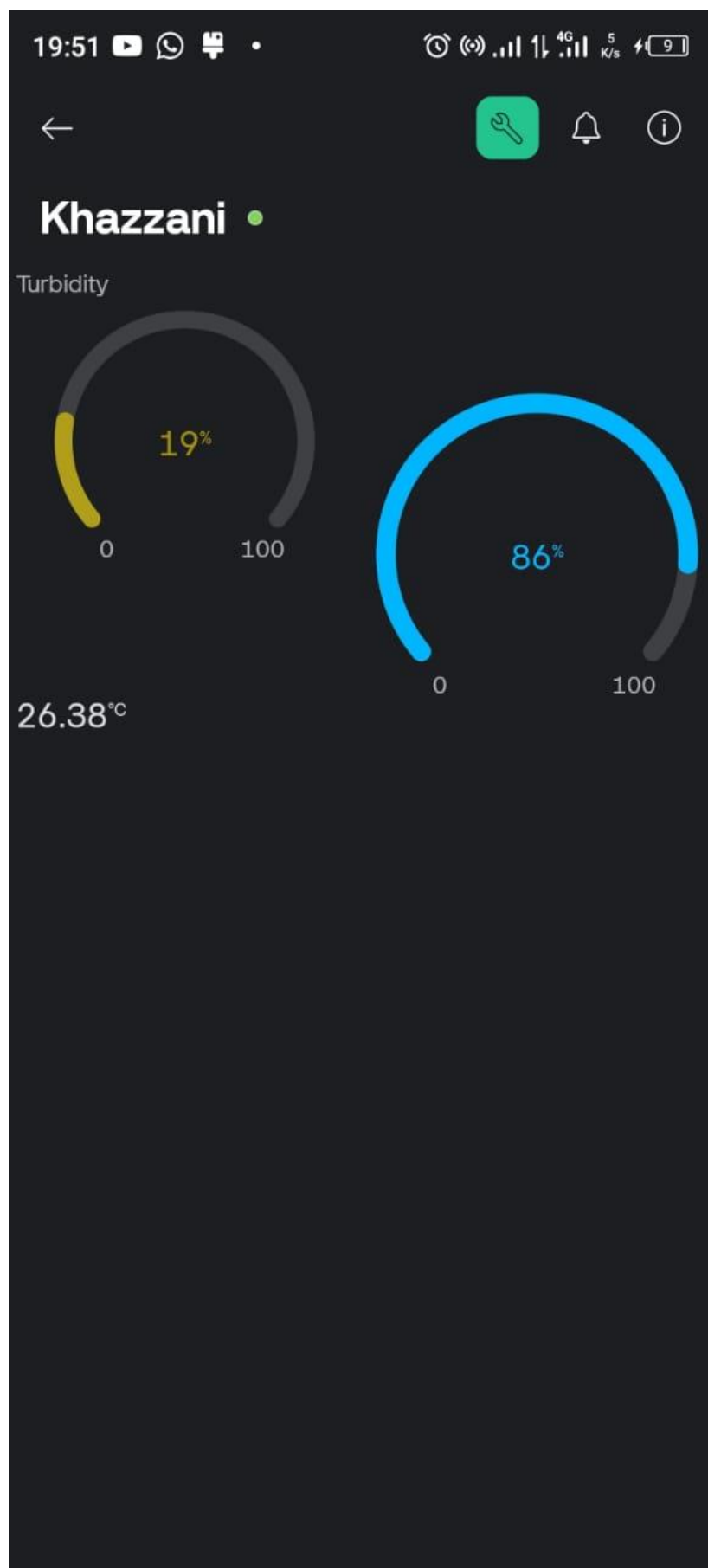
This project is scalable by adding a water pump to empty the tank in case of contamination, or by adding a mini water purification device if it becomes contaminated. In addition to a water heater that heats the water to a certain temperature, which is activated when the water temperature drops.

CH4: METHODOLOGY

• 4.1 Components Connection



- **4.2 Blynk Dashboard**



CH5: CONCLUSIONS

A smart water tank level monitoring and motor pump control system using Internet of Things (IoT) technology is an innovative and efficient way to prevent water waste, detect if the turbidity of the water is high and calculate water temperature. This system allows remote monitoring and management of water tank levels through the use of sensors, communication devices, and cloud-based platforms.